

A Cross-Field Template for Reproducible Community Science

Equations, chemical compounds, code, wide figures and exact artifact references

GenesisL1 Community Publication Templates

Blockchain, computer science, chemistry, biochemistry and biology

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COMMUNITY REVIEW

CROSS-FIELD PROFILE

GL1-CPT-PAPER-3.0

Abstract

Background / Purpose. Reproducible community science requires a manuscript format that can express mathematical models, chemical identities, biological nomenclature, software, and verifiable artifacts without allowing infrastructure language to overwhelm the science.

Methods. A two-column $\text{\LaTeX}$ class was implemented with structured abstracts, numbered equations and compounds, language-aware code boxes, wide floats, numerical citations, and optional content-addressed metadata.

Results. The example compiled with standard $\text{\pdfLaTeX}$ and produced stable references to [equation \(1\)](#), [scheme 1](#), [listing 1](#), and [figure 1](#).

Conclusion. The resulting format supports cross-field scientific writing while keeping interpretation, limitations, provenance, and future infrastructure distinct.

Keywords: community science; reproducibility; chemistry; software; provenance

1 Introduction

Scientific manuscripts frequently connect mathematical analysis, molecular structures, biological observations, software, and computational provenance. The template treats these as compatible components rather than as reasons to impose one rhetorical model on every author.

2 Equations

Numbered equations use standard $\text{\LaTeX}$ semantics. For example, an elementary rate model can be written as

$$v = \frac{V_{\max} [S]}{K_m + [S]}, \quad (1)$$

and cited as [equation \(1\)](#). The equation remains human-readable in source and professionally typeset in the document.

3 Chemistry and biochemistry

Chemical formulae use `mhchem`, for example



Biological names may be written as *Escherichia coli*, genes as *lacZ*, and proteins as LacZ. These lightweight commands can be adapted to the nomenclature rules of a specific discipline.

4 Computer science and blockchain

Code is typeset with the standard `listings` package, so the default build does not require shell escape.

```
1 from hashlib import sha256
2
3 def commit(payload: bytes) -> str:
4     # Return a lowercase SHA-256 commitment.
5     return sha256(payload).hexdigest()
```

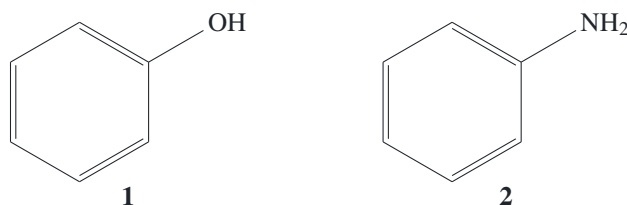
Listing 1. A minimal deterministic transformation

A Solidity fragment can be cited independently:

```
1 pragma solidity ^0.8.24;
2
3 contract ArtifactCommitment {
4     mapping(bytes32 => address) public author;
5
6     function register(bytes32 digest) external {
7         require(author[digest] == address(0), "already
8             registered");
9         author[digest] = msg.sender;
10    }
```

Listing 2. A content commitment stored by a contract

The result in [listing 1](#) describes a byte commitment. It does not establish the quality, legality, safety, or scientific interpretation of the committed object.



Scheme 1. Example structures with persistent compound numbering. Compounds **1** and **2** can be cited consistently throughout the manuscript.

5 Wide figures

Single-column figures use the ordinary `figure` environment. Important diagrams may use `figure*` or the helper environment below to span both columns.

6 Results

The compiled example preserved the order and labels of equations, schemes, code listings, and wide figures. All references were generated by `cleveref`.

EVIDENCE / OBSERVATION

The build produced a two-column paper in which the official GenesisL1 lockup appeared only in the first-page front matter. Subsequent page headers contained text only.

7 Discussion

Our results provide evidence that a single paper class can support the selected fields without making every manuscript about blockchain infrastructure. This observation points towards a broader community publication system, but suitability for a specific journal or reporting guideline must still be evaluated separately.

Interpretation

The main value is not visual uniformity by itself. It is the consistent distinction among direct observations, interpretations, limitations, and exact supporting artifacts.

Limitations and uncertainty

This example is a template demonstration rather than an empirical validation study. Chemical structures generated from SMILES are intentionally left to a separate, reviewable preprocessing step. The default manuscript build consumes committed graphics or `chemfig` source and does not execute external converters.

8 Conclusion

A supportive manuscript container can provide technical precision while leaving scientific judgement, disciplinary conventions, and authorial voice intact.

Artifact record

Provenance and reproducibility

Data	doi:10.xxxx/example-dataset
Code	git:0123456789abcdef
Model	model:v1.2.0
Execution	run:example-2026-08-01
Content hash	sha256:placeholder
IPFS CID	bafy...placeholder
Gateway hint	Optional resolver supplied by the author

Template principle. A supportive container, not a Procrustean bed. Remove modules that do not serve the paper and add field-specific reporting requirements where appropriate.

References

- [1] GenesisL1. *A Sovereign Layer-1 for Verifiable Scientific Infrastructure*. Technical Whitepaper, Version 1.0, 2026.
- [2] GenesisL1 Community Publication Templates. *Scientific Tone and Authoring Guide*. Version 3.0, 2026.

Scientific question → **exact inputs** → **method or model** → **result** → **interpretation**

Each arrow denotes an explicit dependency, not automatic scientific validation.

Figure 1. A full-width figure spanning the two-column scientific body.